

I will formulate this as a reusable theorem-level object: risk measure, control energy, OT target selection, and the resulting optimal control u^* . The cleanest version is deterministic first; stochastic diffusion can be added later as entropy-regularized control.

Thought for 29s >

The clean complete object is:

Choose a terminal distributional risk $\mathcal{R}_y(\nu)$.
 The control u does not directly optimize samples. It chooses a terminal law $\nu = \rho_T^u$.
 The optimal control is the minimum-energy OT geodesic from the source law μ to the best feasible terminal law ν^* .

I will formulate it in the deterministic flow setting first.

1. Controlled transport problem

Let μ be the source distribution, for example Gaussian noise. Let π_y be the ideal conditional target distribution. The controlled generative dynamics are

$$\dot{x}_t = u_t(x_t), \quad x_0 \sim \mu.$$

At the distribution level,

$$\partial_t \rho_t + \nabla \cdot (\rho_t u_t) = 0, \quad \rho_0 = \mu.$$

Define control energy

$$\mathcal{E}_T(u) = \int_0^T \int \|u_t(x)\|^2 d\rho_t(x) dt.$$

By Benamou-Brenier,

$$\inf_{u: \rho_0 = \mu, \rho_T = \nu} \mathcal{E}_T(u) = \frac{1}{T} W_2^2(\mu, \nu).$$

So the control problem reduces to a terminal distribution problem:

choose ν , then move from μ to ν by optimal transport.

That is the key reduction.

2. General risk measure

Take a general terminal task risk

$$\mathcal{R}_y : \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathbb{R}_{\geq 0}.$$

Assume it is L_y -Lipschitz in W_2 :

$$|\mathcal{R}_y(\nu) - \mathcal{R}_y(\tilde{\nu})| \leq L_y W_2(\nu, \tilde{\nu}).$$

This covers risks of the form

$$\mathcal{R}_y(\nu) = \mathbb{E}_{x \sim \nu}[\ell_y(x)]$$

whenever ℓ_y is Lipschitz. It also covers many relaxed constraint losses, reward losses, perceptual losses, measurement losses, and weak distributional risks.

Let the ideal target satisfy

$$\pi_y \in \arg \min_{\nu} \mathcal{R}_y(\nu), \quad \mathcal{R}_y^* = \mathcal{R}_y(\pi_y).$$

Then for any controlled terminal law ρ_T^u ,

$$\mathcal{R}_y(\rho_T^u) - \mathcal{R}_y^* \leq L_y W_2(\rho_T^u, \pi_y).$$

So terminal risk is controlled by how close the controlled terminal distribution is to the ideal conditional target in W_2 .

3. Risk bound induced by the control budget

Assume the control energy is constrained:

$$\mathcal{E}_T(u) \leq B.$$

Then

$$W_2(\mu, \rho_T^u) \leq \sqrt{TB}.$$

Let

$$D_y = W_2(\mu, \pi_y).$$

The best possible terminal law under budget B cannot move farther than radius $\sqrt{TB}$ from μ . Therefore the optimal reachable target is

$$\nu_B^* \in \arg \min_{\nu: W_2(\mu, \nu) \leq \sqrt{TB}} \mathcal{R}_y(\nu).$$

This is the central object.

The induced risk bound is

$$\mathcal{R}_y(\nu_B^*) - \mathcal{R}_y^* \leq L_y \left(D_y - \sqrt{TB} \right)_+.$$

Here

$$(a)_+ = \max(a, 0).$$

Interpretation:

Quantity	Meaning
$D_y = W_2(\mu, \pi_y)$	full transport distance from noise to ideal conditional target
$\sqrt{TB}$	distance reachable under control budget B

Quantity	Meaning
$(D_y - \sqrt{TB})_+$	remaining Wasserstein gap
$L_y(D_y - \sqrt{TB})_+$	worst-case terminal risk gap

So the risk decreases linearly with the reachable OT radius.

4. Optimizing the OT target

The optimal terminal target under budget B is

$$\nu_B^* = \arg \min_{\nu: W_2(\mu, \nu) \leq r_B} \mathcal{R}_y(\nu), \quad r_B = \sqrt{TB}.$$

This is the **risk-constrained OT target selection problem**.

If the only information about $\mathcal{R}_y$ is its W_2 -Lipschitz geometry around π_y , then the optimal robust choice is the point on the Wasserstein geodesic from μ to π_y :

$$\nu_B^* = \rho_{\alpha_B},$$

where

$$\alpha_B = \min \left\{ 1, \frac{\sqrt{TB}}{D_y} \right\}.$$

The geodesic satisfies

$$\rho_0 = \mu, \quad \rho_1 = \pi_y.$$

Thus

$$W_2(\nu_B^*, \pi_y) = (1 - \alpha_B)D_y = (D_y - \sqrt{TB})_+.$$

This is the optimal target distribution under the control budget.

5. Explicit optimal control plan

Assume the optimal transport map from μ to π_y exists:

$$T_y: \mathbb{R}^d \rightarrow \mathbb{R}^d, \quad (T_y)_\# \mu = \pi_y.$$

The budget-optimal terminal target is generated by the partial OT map

$$T_{B,y}(x) = (1 - \alpha_B)x + \alpha_B T_y(x).$$

So

$$\nu_B^* = (T_{B,y})_\# \mu.$$

The optimal controlled trajectory is

$$x_t = \left(1 - \frac{\alpha_B t}{T}\right) x_0 + \frac{\alpha_B t}{T} T_y(x_0).$$

Therefore the optimal open-loop control along each particle path is

$$u_t^*(x_t) = \frac{\alpha_B}{T} (T_y(x_0) - x_0).$$

Equivalently, in particle form:

$$x_T^* = (1 - \alpha_B)x_0 + \alpha_B T_y(x_0),$$

$$u_t^* = \frac{x_T^* - x_0}{T}.$$

So the solution is:

move along the OT geodesic toward the ideal conditional target, but only as far as the budget allows.

6. Hard risk-constrained version

The previous version was compute-constrained:

$$\min_u \mathcal{R}_y(\rho_T^u) \quad \text{s.t.} \quad \mathcal{E}_T(u) \leq B.$$

The risk-constrained version is the dual:

$$\min_u \mathcal{E}_T(u) \quad \text{s.t.} \quad \mathcal{R}_y(\rho_T^u) - \mathcal{R}_y^* \leq \varepsilon.$$

Using the Lipschitz risk bound, it is sufficient to impose

$$W_2(\rho_T^u, \pi_y) \leq \frac{\varepsilon}{L_y}.$$

Let

$$r_\varepsilon = D_y - \frac{\varepsilon}{L_y}.$$

Then the minimum required OT displacement from μ is

$$r_\varepsilon^+ = \left(D_y - \frac{\varepsilon}{L_y} \right)_+.$$

So the optimal terminal target is again the Wasserstein geodesic point

$$\nu_\varepsilon^* = \rho_{\alpha_\varepsilon},$$

with

$$\alpha_\varepsilon = \left(1 - \frac{\varepsilon}{L_y D_y} \right)_+.$$

The corresponding optimal control is

$$u_t^* = \frac{\alpha_\varepsilon}{T} (T_y(x_0) - x_0).$$

The minimum required energy is

$$\mathcal{E}_T(u^*) = \frac{\alpha_\varepsilon^2}{T} D_y^2 = \frac{1}{T} \left(D_y - \frac{\varepsilon}{L_y} \right)_+^2.$$

So the risk-constrained theorem is:

$$\boxed{\min_{u: \mathcal{R}_y(\rho_T^u) - \mathcal{R}_y^* \leq \varepsilon} \mathcal{E}_T(u) \leq \frac{1}{T} \left(D_y - \frac{\varepsilon}{L_y} \right)_+^2}$$

and the minimizing plan is the partial OT geodesic toward π_y .

7. Special case: Wasserstein risk

If the risk itself is

$$\mathcal{R}_y(\nu) = W_2(\nu, \pi_y),$$

then $L_y = 1$, and the result becomes exact:

$$\min_u \mathcal{E}_T(u) \quad \text{s.t.} \quad W_2(\rho_T^u, \pi_y) \leq \varepsilon.$$

The optimal geodesic fraction is

$$\alpha_\varepsilon = \left(1 - \frac{\varepsilon}{D_y} \right)_+.$$

The optimal control is

$$u_t^* = \frac{1}{T} \left(1 - \frac{\varepsilon}{D_y} \right)_+ (T_y(x_0) - x_0).$$

The optimal energy is

$$\mathcal{E}_T(u^*) = \frac{1}{T} (D_y - \varepsilon)_+^2.$$

This is the cleanest exact theorem.

8. With a pretrained diffusion or flow model

Now insert a nominal generative vector field:

$$\dot{x}_t = v_\theta(x_t, t) + u_t(x_t).$$

Let Φ_T^θ be the uncontrolled flow map, and let

$$\bar{\mu}_T = (\Phi_T^\theta)_\# \mu$$

be the terminal distribution of the pretrained sampler.

Now the residual control should not transport from μ to π_y . It should transport from the pretrained terminal law $\bar{\mu}_T$ to the best risk-feasible target.

The terminal target selection becomes

$$\nu_B^* = \arg \min_{\nu: W_2(\bar{\mu}_T, \nu) \leq r_B} \mathcal{R}_y(\nu).$$

For Wasserstein risk,

$$\nu_B^* = \text{geodesic point from } \bar{\mu}_T \text{ to } \pi_y.$$

If

$$\bar{D}_y = W_2(\bar{\mu}_T, \pi_y),$$

then

$$\alpha_B = \min \left\{ 1, \frac{\sqrt{TB}}{\bar{D}_y} \right\}.$$

The target is

$$\nu_B^* = \rho_{\alpha_B}^{\bar{\mu}_T \rightarrow \pi_y}.$$

The residual control u_t^* is the minimum-energy correction that steers the pretrained flow terminal law from $\bar{\mu}_T$ toward ν_B^* .

In infinitesimal form, it solves

$$\begin{aligned} \partial_t \rho_t + \nabla \cdot (\rho_t(v_\theta + u_t)) &= 0, \\ u^* &= \arg \min_u \int_0^T \int \|u_t(x)\|^2 d\rho_t(x) dt \quad \text{s.t.} \quad \rho_T = \nu_B^*. \end{aligned}$$

The Hamilton-Jacobi form is

$$u_t^*(x) = \frac{1}{2} \nabla \phi_t(x),$$

with

$$\partial_t \phi_t + v_\theta(x, t) \cdot \nabla \phi_t + \frac{1}{4} \|\nabla \phi_t\|^2 = 0,$$

and terminal condition determined by the OT potential from ρ_T to ν_B^* .

9. Practical algorithm

For a batch sampler:

1. Draw source particles

$$x_0^i \sim \mu.$$

2. Run the pretrained flow to get terminal particles

$$\bar{x}_T^i = \Phi_T^\theta(x_0^i).$$

3. Construct target particles $y^j \sim \pi_y$, or approximate them using constraint-satisfying samples, reward-weighted samples, posterior samples, or projected samples.
4. Compute empirical OT matching

$$\sigma^* = \arg \min_{\sigma} \frac{1}{n} \sum_{i=1}^n \|\bar{x}_T^i - y^{\sigma(i)}\|^2.$$

5. Compute the budget fraction

$$\alpha_B = \min \left\{ 1, \frac{\sqrt{TB}}{D_y} \right\}.$$

6. Define the optimized terminal target

$$x_{T,B}^{i,*} = (1 - \alpha_B) \bar{x}_T^i + \alpha_B y^{\sigma^*(i)}.$$

7. Optimize the residual control sequence $u_{0:K-1}$ so that the controlled sampler reaches $x_{T,B}^{i,*}$:

$$\min_{u_{0:K-1}} \frac{1}{n} \sum_i \|x_K^{i,u} - x_{T,B}^{i,*}\|^2 + \lambda \sum_{k=0}^{K-1} \frac{1}{n} \sum_i \|u_k(x_k^i)\|^2.$$

This is direct planning, not policy learning.

10. Final compact theorem

Let $\mathcal{R}_y$ be L_y -Lipschitz in W_2 , and let π_y minimize $\mathcal{R}_y$. Let $D_y = W_2(\mu, \pi_y)$. For the controlled transport equation

$$\partial_t \rho_t + \nabla \cdot (\rho_t u_t) = 0, \quad \rho_0 = \mu,$$

under energy budget

$$\mathcal{E}_T(u) \leq B,$$

the optimal robust OT target is

$$\nu_B^* = \rho_{\alpha_B}^{\mu \rightarrow \pi_y}, \quad \alpha_B = \min \left\{ 1, \frac{\sqrt{TB}}{D_y} \right\}.$$

The corresponding control is

$$u_t^* = \frac{\alpha_B}{T} (T_y(x_0) - x_0),$$

and the terminal risk satisfies

$$\mathcal{R}_y(\rho_T^{u^*}) - \mathcal{R}_y^* \leq L_y (D_y - \sqrt{TB})_+.$$

For the hard risk constraint

$$\mathcal{R}_y(\rho_T^u) - \mathcal{R}_y^* \leq \varepsilon,$$

the sufficient optimal geodesic fraction is

$$\alpha_\varepsilon = \left(1 - \frac{\varepsilon}{L_y D_y} \right)_+,$$

and the required energy is

$$\mathcal{E}_T(u^*) = \frac{1}{T} \left(D_y - \frac{\varepsilon}{L_y} \right)_+^2.$$

That is the complete risk-constrained OT control plan.



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look at this result, why could you give similar complete story framing
just change to different risk measure

Thought for 16s >

Yes. The previous result is modular. Its core is not KL. Its core is:

choose a certified risk $R(u) \leq Q(u)$, then optimize the control u against Q .

The uploaded result uses terminal NLL/KL risk and obtains Pareto domination because every baseline is a feasible schedule/grid pair $u_b = (\rho_b, m_b)$, while the proposed scheduler minimizes the same certificate over the full admissible class.  Pasted markdown The same structure works for Wasserstein, IPM, task loss, constraint violation, or posterior risk, provided the bound decomposes into local difficulty and discretization terms.

Here is the complete Wasserstein-risk version.

1. Control class

Let $r \in [0, B]$ be intrinsic noise time. A generative control is

$$u = (\rho, m, \nu),$$

where

$$\begin{aligned} \rho(r) &= \frac{dt}{dr}, & \int_0^B \rho(r) dr &= 1, \\ m(r) &> 0, & \int_0^B m(r) dr &= N, \end{aligned}$$

and ν is the optimized terminal transport target.

Interpretation:

ρ = training or denoising schedule,

m = sampling grid density,

ν = terminal law chosen under a risk constraint.

Every fixed baseline is a special point

$$u_b = (\rho_b, m_b, \nu_b).$$

Cosine, linear, Karras, log-SNR grids, OT-CFM paths, and Fisher-information schedules are fixed choices inside this class, as in the uploaded theorem.  Pasted markdown

2. Choose Wasserstein terminal risk

Let π_y be the ideal conditional target distribution. Use

$$R_W(u) = W_2^2(\rho_T^u, \pi_y).$$

The terminal law ρ_T^u is produced by the learned sampler under schedule ρ , grid m , and terminal target ν .

Assume the certified risk bound has the form

$$R_W(u) \leq Q_W(u),$$

where

$$Q_W(u) = Q_0 + W_2^2(\nu, \pi_y) + \frac{c_{\text{stat}}}{n} \int_0^B \frac{a_W(r)}{\rho(r)} dr + c_{\text{disc}} \int_0^B \frac{d_W(r)}{m(r)^p} dr.$$

Meaning:

Q_0 = irreducible model or approximation floor,

$W_2^2(\nu, \pi_y)$ = terminal target mismatch,

$a_W(r)$ = local Wasserstein statistical difficulty,

$d_W(r)$ = local Wasserstein discretization stiffness,

p = solver order.

This is the Wasserstein analogue of the uploaded KL certificate

$$Q(u) = Q_0 + \frac{c_{\text{stat}}}{n} \int_0^B \frac{a(r)}{\rho(r)} dr + c_{\text{disc}} \int_0^B \frac{d(r)}{m(r)^p} dr.$$

The proof structure is identical; only a, d, Q become a_W, d_W, Q_W . The uploaded result already proves the optimizer form for the statistical and discretization terms. Pasted markdown

3. Risk-constrained objective

The accuracy-constrained efficient control problem is

$$u^*(\varepsilon) = \arg \min_u N(u)$$

subject to

$$Q_W(u) \leq \varepsilon.$$

Equivalently,

$$\min_{\rho, m, \nu} N$$

subject to

$$Q_0 + W_2^2(\nu, \pi_y) + \frac{c_{\text{stat}}}{n} \int_0^B \frac{a_W(r)}{\rho(r)} dr + c_{\text{disc}} \int_0^B \frac{d_W(r)}{m(r)^p} dr \leq \varepsilon.$$

This is the complete risk-constrained Wasserstein scheduler.

4. Optimize the terminal OT target

Let the source distribution be μ , for example Gaussian noise or the terminal law of an uncontrolled pretrained sampler. Let

$$D_y = W_2(\mu, \pi_y).$$

Suppose the control energy budget for terminal transport is B_{ctrl} . By Benamou-Brenier,

$$\inf_{u: \rho_0 = \mu, \rho_T = \nu} \int_0^T \mathbb{E}_{\rho_t} \|u_t\|^2 dt = \frac{1}{T} W_2^2(\mu, \nu).$$

Thus the reachable Wasserstein radius is

$$r_{\text{ctrl}} = \sqrt{T B_{\text{ctrl}}}.$$

The optimized terminal target is

$$\nu^* = \arg \min_{\nu: W_2(\mu, \nu) \leq r_{\text{ctrl}}} W_2^2(\nu, \pi_y).$$

The solution is the Wasserstein geodesic point from μ to π_y :

$$\nu^* = \rho_{\alpha}^{\mu \rightarrow \pi_y},$$

where

$$\alpha = \min \left\{ 1, \frac{r_{\text{ctrl}}}{D_y} \right\}.$$

Therefore

$$W_2(\nu^*, \pi_y) = (D_y - r_{\text{ctrl}})_+.$$

So the terminal target contribution becomes

$$W_2^2(\nu^*, \pi_y) = (D_y - \sqrt{T B_{\text{ctrl}}})_+^2.$$

This is the Wasserstein-risk replacement for terminal KL/NLL.

5. Optimize the statistical schedule

Solve

$$\min_{\rho} \int_0^B \frac{a_W(r)}{\rho(r)} dr$$

subject to

$$\int_0^B \rho(r) dr = 1.$$

The optimizer is

$$\rho_W^*(r) = \frac{\sqrt{a_W(r)}}{\int_0^B \sqrt{a_W(s)} ds}.$$

The minimized statistical term is

$$B_{\text{stat},W}^* = \frac{c_{\text{stat}}}{n} \left(\int_0^B \sqrt{a_W(r)} dr \right)^2.$$

So Wasserstein-risk training should spend more resolution where the local score or velocity error has the largest amplified effect on final W_2 error.

6. Optimize the sampling grid

Solve

$$\min_m \int_0^B \frac{d_W(r)}{m(r)^p} dr$$

subject to

$$\int_0^B m(r) dr = N.$$

The optimizer is

$$m_W^*(r) = \frac{N d_W(r)^{1/(p+1)}}{\int_0^B d_W(s)^{1/(p+1)} ds}.$$

The optimal local step size is

$$\Delta r_W^*(r) \propto d_W(r)^{-1/(p+1)}.$$

The minimized discretization certificate is

$$B_{\text{disc},W}^*(N) = c_{\text{disc}} \frac{\left(\int_0^B d_W(r)^{1/(p+1)} dr \right)^{p+1}}{N^p}.$$

7. Final optimized certificate

After optimizing ν, ρ, m , the best Wasserstein certificate is

$$Q_W^*(N, B_{\text{ctrl}}) = Q_0 + (D_y - \sqrt{TB_{\text{ctrl}}})_+^2 + \frac{c_{\text{stat}}}{n} \left(\int_0^B \sqrt{a_W(r)} dr \right)^2 + c_{\text{disc}} \frac{\left(\int_0^B d_W(r)^{1/(p+1)} dr \right)^{p+1}}{N^p}.$$

This is the full Wasserstein-risk analogue of the KL result.

8. Minimum NFE under Wasserstein accuracy

Given target accuracy ε , the minimum NFE is

$$N_W^*(\varepsilon) = \left\lceil \left[\frac{c_{\text{disc}} \left(\int_0^B d_W(r)^{1/(p+1)} dr \right)^{p+1}}{\varepsilon - Q_0 - (D_y - \sqrt{TB_{\text{ctrl}}})_+^2 - \frac{c_{\text{stat}}}{n} \left(\int_0^B \sqrt{a_W(r)} dr \right)^2} \right]^{1/p} \right\rceil.$$

This is valid when the denominator is positive. If it is nonpositive, the requested Wasserstein accuracy cannot be certified at the given model, data, and control budget.

9. Pareto domination theorem

Let $u_b = (\rho_b, m_b, \nu_b)$ be any baseline with NFE N_b . Then

$$Q_W(u_b) \geq Q_W^*(N_b, B_{\text{ctrl}}).$$

Set

$$\varepsilon = Q_W(u_b).$$

By definition,

$$N_W^*(\varepsilon) = \min\{N : Q_W^*(N, B_{\text{ctrl}}) \leq \varepsilon\}.$$

Since

$$Q_W^*(N_b, B_{\text{ctrl}}) \leq Q_W(u_b) = \varepsilon,$$

we get

$$N_W^*(\varepsilon) \leq N_b.$$

Also,

$$R_W(u^*) \leq Q_W(u^*) \leq Q_W(u_b).$$

Therefore,

$$\boxed{R_W(u^*) \leq Q_W(u_b), \quad N(u^*) \leq N(u_b).}$$

Thus the Wasserstein-risk scheduler weakly Pareto-dominates every fixed baseline under the same certified Wasserstein accuracy criterion.

10. Final theorem statement

Theorem. Let $\mathcal{U}$ contain baseline schedules, sampling grids, and terminal transport targets. Suppose squared Wasserstein terminal risk is certified by

$$R_W(u) = W_2^2(\rho_T^u, \pi_y) \leq Q_W(u),$$

with

$$Q_W(u) = Q_0 + W_2^2(\nu, \pi_y) + \frac{c_{\text{stat}}}{n} \int_0^B \frac{a_W(r)}{\rho(r)} dr + c_{\text{disc}} \int_0^B \frac{d_W(r)}{m(r)^p} dr.$$

Then the optimal risk-constrained control is

$$\nu^* = \rho_\alpha^{\mu \rightarrow \pi_y}, \quad \alpha = \min \left\{ 1, \frac{\sqrt{TB_{\text{ctrl}}}}{W_2(\mu, \pi_y)} \right\},$$

$$\rho_W^*(r) \propto \sqrt{a_W(r)},$$

$$m_W^*(r) \propto d_W(r)^{1/(p+1)}.$$

For every baseline $u_b \in \mathcal{U}$, choosing $\varepsilon = Q_W(u_b)$ yields a controlled sampler $u^*(\varepsilon)$ such that

$$R_W(u^*) \leq Q_W(u^*) \leq Q_W(u_b),$$

and

$$N(u^*) \leq N(u_b).$$

This is the same complete story as the KL scheduler, but with a different certified risk. The theorem does not depend on KL. It depends on the existence of a decomposable risk certificate.



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Q_0 = irreducible model or approximation floor,
 $W_2(v, \pi_y)$ = terminal target mismatch, $W_2^2(\nu, \pi_y)$ = terminal target mismatch,
 $a_W(r)$ = local Wasserstein statistical difficulty, $d_W(r)$ = local Wasserstein statisti...

is this notation concrete, i don't think so, could you make them concrete and reexpress the result

The weak point is exactly the labels Q_0, a_W, d_W . I will replace them by quantities computed from vector-field error, terminal amplification, and solver local truncation error.

Thought for 24s >

Yes. The previous notation was too abstract. The correct way is to define $Q_0, a_W(r)$, and $d_W(r)$ from **actual vector-field error propagation**.

Below is the concrete Wasserstein version of the result. It keeps the same optimizer structure as the uploaded scheduler theorem, where schedules are represented by $\rho(r)$, grids by $m(r)$, and the certificate is optimized over the admissible class. Pasted markdown

1. Concrete controlled flow setting

Let $r \in [0, B]$ be intrinsic noise time. Let the ideal conditional flow be

$$\frac{dx_r}{dr} = v_y^*(x_r, r), \quad x_0 \sim \mu, \quad x_B \sim \pi_y.$$

Here:

$$\mu = \text{noise prior}, \quad \pi_y = \text{ideal conditional target distribution}.$$

A learned sampler uses

$$\frac{dx_r}{dr} = \hat{v}_\rho(x_r, r),$$

where $\rho(r)$ is the training or schedule density:

$$\rho(r) > 0, \quad \int_0^B \rho(r) dr = 1.$$

The numerical solver uses a step-density

$$m(r) > 0, \quad \int_0^B m(r) dr = N.$$

Large $m(r)$ means more solver steps around noise level r .

The terminal Wasserstein risk is

$$R_W(\rho, m) = W_2^2(\hat{\rho}_B^{\rho, m}, \pi_y),$$

where $\hat{\rho}_B^{\rho, m}$ is the terminal law produced by the learned and discretized sampler.

2. Concrete quantities

Assume the ideal vector field has Lipschitz constant

$$\|v_y^*(x, r) - v_y^*(x', r)\| \leq L_y(r) \|x - x'\|.$$

Define terminal amplification from time r to B :

$$A_y(r) = \exp \left(\int_r^B L_y(s) ds \right).$$

This is concrete: an error in the vector field at time r is amplified by at most $A_y(r)$ at the terminal distribution.

Now define the local statistical estimation difficulty:

$$\sigma_y^2(r) = \mathbb{E}_{x \sim \rho_r^*} [\text{Var}(\text{training target for } v_y^*(x, r) \mid x, r)].$$

If n training samples are allocated according to $\rho(r)$, then the local learned vector-field error scales as

$$\mathbb{E}_{x \sim \rho_r^*} \|\hat{v}_\rho(x, r) - v_y^*(x, r)\|^2 \lesssim \frac{\sigma_y^2(r)}{n\rho(r)} + b_y^2(r),$$

where

$$b_y^2(r) = \inf_{v \in \mathcal{F}} \mathbb{E}_{x \sim \rho_r^*} \|v(x, r) - v_y^*(x, r)\|^2$$

is the local approximation bias of the model class.

For the numerical solver, let

$$\tau_y(r)$$

be the local truncation-error coefficient of the solver at time r . For example, for Euler, $\tau_y(r)$ is controlled by derivatives of v_y^* , roughly by quantities like $\|\partial_r v_y^*\|$, $\|\nabla v_y^* v_y^*\|$, and local stiffness.

Now define:

$$\begin{aligned} Q_0 &= B \int_0^B A_y(r)^2 b_y^2(r) dr, \\ a_W(r) &= B A_y(r)^2 \sigma_y^2(r), \\ d_W(r) &= A_y(r)^2 \tau_y(r)^2. \end{aligned}$$

These are no longer placeholders.

They mean:

$$\begin{aligned} Q_0 &= \text{amplified model-class bias,} \\ a_W(r) &= \text{amplified local vector-field estimation variance,} \\ d_W(r) &= \text{amplified local solver truncation error.} \end{aligned}$$

3. Concrete Wasserstein risk bound

By flow stability,

$$W_2(\hat{\rho}_B, \pi_y) \lesssim \int_0^B A_y(r) (\mathbb{E}_{\rho_r^*} \|\hat{v}_\rho - v_y^*\|^2)^{1/2} dr + \text{solver error.}$$

Using Jensen/Cauchy bounds, this gives the squared certificate

$$R_W(\rho, m) \leq Q_W(\rho, m),$$

with

$$Q_W(\rho, m) = Q_0 + \frac{1}{n} \int_0^B \frac{a_W(r)}{\rho(r)} dr + c_{\text{disc}} \int_0^B \frac{d_W(r)}{m(r)^p} dr.$$

Substituting the concrete definitions:

$$Q_W(\rho, m) = B \int_0^B A_y(r)^2 b_y^2(r) dr + \frac{B}{n} \int_0^B \frac{A_y(r)^2 \sigma_y^2(r)}{\rho(r)} dr + c_{\text{disc}} \int_0^B \frac{A_y(r)^2 \tau_y(r)^2}{m(r)^p} dr.$$

This is the concrete certificate.

4. Optimize the schedule

We solve

$$\min_{\rho} \int_0^B \frac{A_y(r)^2 \sigma_y^2(r)}{\rho(r)} dr$$

subject to

$$\int_0^B \rho(r) dr = 1.$$

The solution is

$$\rho_W^*(r) = \frac{A_y(r) \sigma_y(r)}{\int_0^B A_y(s) \sigma_y(s) ds}.$$

So the optimal training or schedule density is:

$$\boxed{\rho_W^*(r) \propto A_y(r) \sigma_y(r).}$$

Interpretation:

Spend more schedule mass where local vector-field estimation is hard and where local errors strongly

affect the terminal distribution.

Equivalently,

$$\beta_W^*(t) = \frac{dr}{dt} \propto \frac{1}{A_y(r(t))\sigma_y(r(t))}.$$

So the process slows down where $A_y(r)\sigma_y(r)$ is large.

5. Optimize the sampling grid

We solve

$$\min_m \int_0^B \frac{A_y(r)^2 \tau_y(r)^2}{m(r)^p} dr$$

subject to

$$\int_0^B m(r) dr = N.$$

The solution is

$$m_W^*(r) = \frac{N \left(A_y(r)^2 \tau_y(r)^2 \right)^{1/(p+1)}}{\int_0^B \left(A_y(s)^2 \tau_y(s)^2 \right)^{1/(p+1)} ds}.$$

Equivalently,

$$m_W^*(r) \propto \left(A_y(r) \tau_y(r) \right)^{2/(p+1)}.$$

Thus the optimal local step size is

$$\Delta r_W^*(r) \propto \left(A_y(r) \tau_y(r) \right)^{-2/(p+1)}.$$

For Euler with $p = 1$,

$$\Delta r_W^*(r) \propto \frac{1}{A_y(r) \tau_y(r)}.$$

Interpretation:

Put more solver steps where the dynamics are locally numerically stiff and where local numerical error is strongly terminal-amplified.

6. Optimized Wasserstein certificate

Plugging the optimal ρ_W^* and m_W^* into the bound gives

$$Q_W^*(N) = B \int_0^B A_y(r)^2 b_y^2(r) dr + \frac{B}{n} \left(\int_0^B A_y(r) \sigma_y(r) dr \right)^2 + c_{\text{disc}} \frac{\left(\int_0^B \left(A_y(r)^2 \tau_y(r)^2 \right)^{1/(p+1)} dr \right)^{p+1}}{N^p}.$$

This is the concrete risk certificate.

7. Minimum NFE for target Wasserstein accuracy

Given target accuracy

$$R_W(\rho, m) \leq \varepsilon,$$

it is sufficient to require

$$Q_W^*(N) \leq \varepsilon.$$

Therefore

$$N_W^*(\varepsilon) = \left\lceil \left[\frac{c_{\text{disc}} \left(\int_0^B \left(A_y(r)^2 \tau_y(r)^2 \right)^{1/(p+1)} dr \right)^{p+1}}{\varepsilon - B \int_0^B A_y(r)^2 b_y^2(r) dr - \frac{B}{n} \left(\int_0^B A_y(r) \sigma_y(r) dr \right)^2} \right]^{1/p} \right\rceil.$$

This is concrete: the required NFE depends on three measurable profiles:

$$A_y(r), \quad \sigma_y(r), \quad \tau_y(r).$$

8. Add risk-constrained OT target selection

The above assumes the ideal target is exactly π_y . If we also allow the terminal OT target ν to be optimized under control budget B_{ctrl} , then define

$$D_y = W_2(\mu, \pi_y), \quad r_{\text{ctrl}} = \sqrt{TB_{\text{ctrl}}}.$$

The best reachable terminal target is

$$\nu^* = \rho_{\alpha}^{\mu \rightarrow \pi_y},$$

where

$$\alpha = \min \left\{ 1, \frac{r_{\text{ctrl}}}{D_y} \right\}.$$

Then

$$W_2^2(\nu^*, \pi_y) = (D_y - r_{\text{ctrl}})_+^2.$$

The full certificate becomes

$$Q_W^*(N, B_{\text{ctrl}}) = (D_y - \sqrt{TB_{\text{ctrl}}})_+^2 + B \int_0^B A_y(r)^2 b_y^2(r) dr + \frac{B}{n} \left(\int_0^B A_y(r) \sigma_y(r) dr \right)^2 + c_{\text{disc}} \frac{\left(\int_0^B (A_y(r)^2 \tau_y(r)^2) \right)^1}{N^p}$$

This is the complete concrete expression.

9. Final result, re-expressed

The optimal Wasserstein-risk constrained scheduler is:

$$\nu^* = \rho_{\alpha}^{\mu \rightarrow \pi_y}, \quad \alpha = \min \left\{ 1, \frac{\sqrt{TB_{\text{ctrl}}}}{W_2(\mu, \pi_y)} \right\}$$

$$\rho_W^*(r) = \frac{A_y(r) \sigma_y(r)}{\int_0^B A_y(s) \sigma_y(s) ds}$$

$$m_W^*(r) = \frac{N (A_y(r)^2 \tau_y(r)^2)^{1/(p+1)}}{\int_0^B (A_y(s)^2 \tau_y(s)^2)^{1/(p+1)} ds}$$

where

$$A_y(r) = \exp \left(\int_r^B L_y(s) ds \right),$$

$$\sigma_y^2(r) = \text{local conditional vector-field estimation variance,}$$

$$\tau_y(r) = \text{local solver truncation-error coefficient.}$$

So the theorem is no longer “optimize abstract a_W, d_W .” It is:

The Wasserstein-optimal schedule allocates training/sampling time proportional to terminal-amplified vector-field uncertainty, and allocates solver steps proportional to terminal-amplified numerical stiffness.

The dominance statement then follows exactly as in the uploaded result: every baseline is a fixed (ρ_b, m_b, ν_b) , while the proposed method optimizes the same certified Wasserstein frontier.

